



BOARD QUESTION PAPER : MARCH 2016

Notes:

- All questions are compulsory.
- Figures to the right indicate full marks.
- Answer to every question must be written on a new page.
- L.P.P. problem should be solved on graph paper.
- Log table will be provided on request.
- Write answers of Section – I and Section – II in one answer book.

Section – I**Q.1. Attempt any SIX of the following:****[12]**

- If $y = (\sin x)^x$, find $\frac{dy}{dx}$. (2)
- If $A = \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$ show that $A^2 - 2A$ is a scalar matrix. (2)
- Write the negation of the following statements:
 - $\forall y \in \mathbb{N}, y^2 + 3 \leq 7$
 - If the lines are parallel then their slopes are equal. (2)
- The total revenue $R = 720x - 3x^2$ where x is number of items sold. Find x for which total revenue R is increasing. (2)
- Evaluate: $\int \frac{\sec^2 x}{\tan^2 x + 4} dx$ (2)
- Find $\frac{dy}{dx}$, if $y = \cos^{-1}(\sin 5x)$ (2)
- Discuss the continuity of function f at $x = 0$
Where $f(x) = \frac{\sqrt{4+x}-2}{3x}$, for $x \neq 0$
$$= \frac{1}{12}, \text{ for } x = 0$$
 (2)
- State which of the following sentences are statements. In case of statement, write down the truth value:
 - Every quadratic equation has only real roots.
 - $\sqrt{-4}$ is a rational number. (2)

Q.2. (A) Attempt any TWO of the following:**[6][14]**

- Solve the following equations by the inversion method:
 $2x + 3y = -5$ and $3x + y = 3$ (3)
- Find x and y , if $\left\{ 3 \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 5 & -2 \\ -3 & -4 & 4 \end{bmatrix} \right\} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$ (3)
- Evaluate: $\int \tan^{-1} x dx$. (3)

**(B) Attempt any TWO of the following:****[8]**

- i. (a) Express the truth of each of the following statements using Venn diagram.
- (1) All teachers are scholars and scholars are teachers.
 - (2) If a quadrilateral is a rhombus then it is a parallelogram.
- (b) Write converse and inverse of the following statement:
“If Ravi is good in logic then Ravi is good in Mathematics.” (4)
- ii. Find the area of the region bounded by the lines $2y + x = 8$, $x = 2$ and $x = 4$. (4)
- iii. Evaluate: $\int_3^9 \frac{\sqrt[3]{12-x}}{\sqrt[3]{x} + \sqrt[3]{12-x}} dx$ (4)

Q.3. (A) Attempt any TWO of the following:**[6][14]**

- i. If $f(x) = \frac{e^{2x} - 1}{ax}$, for $x < 0$, $a \neq 0$
 $= 1$, for $x = 0$
 $= \frac{\log(1+7x)}{bx}$, for $x > 0$, $b \neq 0$
- Is continuous at $x = 0$ then find a and b . (3)
- ii. If the function f is continuous at $x = 0$, then find $f(0)$
 where $f(x) = \frac{\cos 3x - \cos x}{x^2}$, $x \neq 0$ (3)
- iii. If $f'(x) = 4x^3 - 3x^2 + 2x + k$ and $f(0) = 1$, $f(1) = 4$, find $f(x)$. (3)

(B) Attempt any TWO of the following:**[8]**

- i. Find MPC (Marginal Propensity to Consume) and APC (Average Propensity to Consume) if the expenditure E_c of a person with income I is given as $E_c = (0.0003) I^2 + (0.075) I$ when $I = 1000$. (4)
- ii. Cost of assembling x wallclocks is $\left(\frac{x^3}{3} - 40x^2\right)$ and labour charges are $500x$. Find the number of wallclocks to be manufactured for which marginal cost is minimum. (4)
- iii. If $\cos^{-1} \left(\frac{x^2 - y^2}{x^2 + y^2} \right) = 2k$,
 show that $y \frac{dy}{dx} = x \tan^2 k$. (4)

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Section – I

Question 1 to 3 (based on section I) are given in our book **STD XII (COMMERCE) MATHEMATICS AND STATISTICS - I**

Section – II**Q.4. Attempt any SIX of the following:****[12]**

- Anandi and Rutuja invested ₹ 10,000 each in a business. Anandi withdrew her capital after 7 months. Rutuja continued for the year. After one year, the profit earned by them was ₹ 5,700. Find the profit earned by each person. (2)
- Calculate age specific death (A-SDR) rates for the following data:

Age group (in years)	Population ('000)	Number of Deaths
Below 10	25	50
10 – 30	30	90
30 – 45	40	160
45 – 70	20	100

(2)

- For a bivariate data $b_{YX} = -1.2$ and $b_{XY} = -0.3$, find the correlation coefficient between x and y . (2)
- A random variable x has the following probability distribution:

x	0	1	2	3	4	5	6
$P(X=x)$	k	$3k$	$5k$	$7k$	$9k$	$11k$	$13k$

Find 'k'.

(2)

- The probability distribution function of continuous random variable X is given by

$$f(x) = \frac{x}{4}, 0 < x < 2$$

$$= 0, \text{ otherwise}$$

Find $P(x \leq 1)$.

(2)

- From the two regression equations $y = 4x - 5$ and $3x = 2y + 5$ find $\bar{x}$ and $\bar{y}$. (2)

- Draw scatter diagram for the following data and interpret it:

x	10	20	30	40	50	60	70
y	32	20	24	36	40	28	38

(2)

- If $\Sigma d^2 = 66$ and $n = 10$ then find the rank correlation coefficient. (2)

Q.5. (A) Attempt any TWO of the following:**(6)[14]**

- Determine l_{92} and l_{93} , given that $l_{91} = 97$, $d_{91} = 38$ and $q_{92} = \frac{27}{59}$. (3)



- ii. Calculate CDR for districts A and B and compare them.
Also state which district is more healthy.

(3)

Age group (in years)	District A		District B	
	No. of Persons ('000)	No. of Deaths	No. of Persons ('000)	No. of Deaths
0 – 15	1	20	2	50
15 – 60	3	30	7	70
60 and above	2	40	1	25

- iii. If for a bivariate data $\bar{x} = 10$, $\bar{y} = 12$, $\text{Var}(X) = 9$, $\sigma_Y = 4$ and $r = 0.6$, estimate y when $x = 5$.

(3)

(B) Attempt any TWO of the following:

(8)

- i. Calculate the coefficient of correlation between X and Y series from the following data:

$$n = 15, \bar{x} = 25, \bar{y} = 18, \sigma_X = 3.01, \sigma_Y = 3.03, \sum (x_i - \bar{x})(y_i - \bar{y}) = 122$$

(4)

- ii. Solve the following minimal assignment problem and hence find minimum time where ‘–’ indicates that job cannot be assigned to the machine:

Machines	Processing time in hours				
	A	B	C	D	E
M ₁	9	11	15	10	11
M ₂	12	9	–	10	9
M ₃	–	11	14	11	7
M ₄	14	8	12	7	8

(4)

- iii. Solve the following maximal assignment problem:

Branch Manager	Monthly Business (₹ lakh)			
	A	B	C	D
P	11	11	9	9
Q	13	16	11	10
R	12	17	13	8
S	16	14	16	12

(4)

Q.6. (A) Attempt any TWO of the following:

(6)[14]

- i. Find the true discount, banker's discount and banker's gain on a bill of ₹ 36,600 due 4 months hence at 5% p.a.

(3)

- ii. Mr. Anil wants to invest at most ₹ 60,000 in Fixed Deposit (F.D.) and Public Provident Fund (P.P.F.). He wants to invest at least ₹ 20,000 in F.D. and at least ₹ 15,000 in P.P.F. The rate of interest on F.D. is 8% p.a. and that on P.P.F. is 10% p.a. Formulate the above problem as L.P.P. to determine maximum yearly income.

(3)

- iii. Find graphical solution for the following system of linear inequations:

$$3x + 2y \leq 180; x + 2y \leq 120, x \geq 0, y \geq 0$$

Hence find co-ordinates of corner points of the common region.

(3)

(B) Attempt any TWO of the following:

(8)

- i. Mrs. Menon plans to save for her daughter's marriage. She wants to accumulate a sum of ₹ 4,00,000 at the end of 4 years. How much should she invest at the end of each year from now, if she can get interest compounded at 10% p.a.? [Given : $(1.1)^4 = 1.4641$]

(4)

- ii. A car valued at ₹ 4,00,000 is insured for ₹ 2,50,000. The rate of premium is 5% less 20%. How much loss does the owner bear including the premium if value of the car is reduced to 60% of its original value?

(4)

- iii. If a random variable X has probability distribution function

$$f(x) = \frac{c}{x}, 1 < x < 3, c > 0,$$

find c, E(X) and Var (X).

(4)